

CLASS 8 · CBSE · GANITA PRAKASH PART 1 · SHORT NOTES

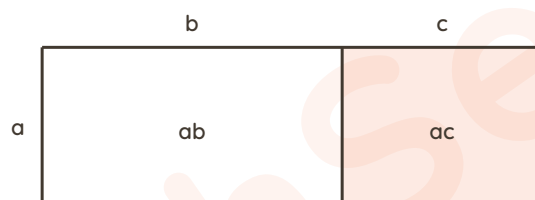
Chapter 6: We Distribute, Yet Things Multiply

Topic-wise short notes with solved examples

The distributive property and what it leads to: multiplying two brackets, the identities for the square of a sum, the square of a difference and the difference of two squares, quick multiplication methods, and describing patterns with algebra.

1 The distributive property

- Multiplication distributes over addition: $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.
- It also works for subtraction, $a(b - c) = ab - ac$, and for any number of terms inside the bracket.
- It holds for all integers and fractions, positive or negative.



A rectangle a by $(b + c)$ has area $ab + ac$

- An **identity** is an equality between two expressions that is true for every value of the letters.
Example: $a(b + 8) = ab + 8a$.
- Only **like terms** (same letters with the same powers) can be added: $a^2b + 2a^2b = 3a^2b$, but $\frac{3}{2}a^2$ and $\frac{3}{10}a$ cannot be combined.

Example 1.1 Expand $\frac{3a}{2} \left(a - b + \frac{1}{5} \right)$.

SOLUTION

Multiply each term by $\frac{3a}{2}$.

$$\frac{3}{2}a^2 - \frac{3}{2}ab + \frac{3}{10}a$$

Example 1.2 By how much does the product 23×27 increase if 27 is increased by 1?

SOLUTION

$$23(27 + 1) = 23 \times 27 + 23$$

By 23

2 Multiplying two brackets

- Multiply each term of the first bracket by each term of the second, then add.

$$(a + m)(b + n) = ab + an + mb + mn$$

- With signs, use the usual rules for integers: $(a + u)(b - v) = ab - av + ub - uv$, $(a - u)(b + v) = ab + av - ub - uv$ and $(a - u)(b - v) = ab - av - ub + uv$.

	b	n
a	ab	an
m	mb	mn

$(a + m)(b + n)$ is the sum of four areas

- How a product changes:** if both numbers go up by 1, $(a + 1)(b + 1) = ab + a + b + 1$, so the product rises by $a + b + 1$. If one goes up by 1 and the other down by 1, $(a + 1)(b - 1) = ab + b - a - 1$.

Example 2.1 Expand $(3 - x)(x - 6)$.

SOLUTION

$$3x - 18 - x^2 + 6x$$

$$-x^2 + 9x - 18$$

Example 2.2 Expand $(4y + 7)(y + 11z - 3)$.

SOLUTION

$$4y^2 + 44yz - 12y + 77z - 21$$

$$4y^2 + 44yz - 5y + 77z - 21$$

Example 2.3 The product of two numbers stays the same when one is increased by 2 and the other is decreased by 4. How are the numbers related?

SOLUTION

$$(a + 2)(b - 4) = ab \text{ gives } ab - 4a + 2b - 8 = ab, \text{ so } 2b = 4a + 8$$

$$b = 2a + 4, \text{ for example } 1 \text{ and } 6, \text{ or } 3 \text{ and } 10$$

Example 2.4 Which is larger: 25×75 or 26×74 ?

SOLUTION

$$26 \times 74 = (25 + 1)(75 - 1) = 25 \times 75 + 75 - 25 - 1 = 25 \times 75 + 49$$

26×74 is larger, by 49

Example 2.5 Expand $(a + b)(a^2 + 2ab + b^2)$.

SOLUTION

$$a^3 + 2a^2b + ab^2 + a^2b + 2ab^2 + b^3$$

$$a^3 + 3a^2b + 3ab^2 + b^3$$

3 Quick multiplication

- **By 11:** $N \times 11 = N \times 10 + N$. Writing from the right: the units digit, then the sum of each pair of neighbouring digits (carrying when needed), then the first digit.
- **By 101:** $N \times 101 = N \times 100 + N$. **By 1001:** $N \times 1000 + N$.
- **By 99, 999:** $N \times 99 = N \times 100 - N$ and $N \times 999 = N \times 1000 - N$.
- Brahmagupta (628 CE) stated the distributive property as "multiplication by parts" and used it for fast multiplication.

Example 3.1 Find 3874×11 in one line.

SOLUTION

From the right: 4 ; $7 + 4 = 11$, write 1 carry 1; $8 + 7 + 1 = 16$, write 6 carry 1; $3 + 8 + 1 = 12$, write 2 carry 1; $3 + 1 = 4$.

42614

Example 3.2 Find 9734×99 .

SOLUTION

$$9734 \times 100 - 9734 = 973400 - 9734$$

963666

Example 3.3 Find 265×1001 .

SOLUTION

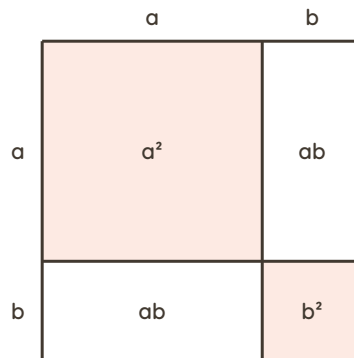
$$265000 + 265$$

265265

4 Square of a sum

$$(a + b)^2 = a^2 + 2ab + b^2$$

- Reason: $(a + b)(a + b) = a^2 + ab + ba + b^2$.
- Picture: a square of side $a + b$ splits into squares of area a^2 and b^2 and two rectangles of area ab each.



$$(a + b)^2 = a^2 + 2ab + b^2$$

- $(a + b)^2$ is not $a^2 + b^2$. The middle term $2ab$ must be included.

Example 4.1 Find 65^2 and 104^2 .

SOLUTION

$$65^2 = (60 + 5)^2 = 3600 + 600 + 25 = 4225$$

$$104^2 = (100 + 4)^2 = 10000 + 800 + 16 = 10816$$

4225 and 10816

Example 4.2 Expand $(6x + 5)^2$.

SOLUTION

$$(6x)^2 + 2(6x)(5) + 5^2$$

$$36x^2 + 60x + 25$$

5 Square of a difference

$$(a - b)^2 = a^2 - 2ab + b^2$$

- It follows from the first identity by writing $a - b$ as $a + (-b)$.
- $(a - b)^2 = (b - a)^2$, since squaring removes the sign.

Example 5.1 Find 99^2 and 58^2 .

SOLUTION

$$99^2 = (100 - 1)^2 = 10000 - 200 + 1 = 9801$$

$$58^2 = (60 - 2)^2 = 3600 - 240 + 4 = 3364$$

9801 and 3364

Example 5.2 Expand $\left(7y - \frac{3}{4}\right)^2$.

SOLUTION

$$(7y)^2 - 2(7y)\left(\frac{3}{4}\right) + \left(\frac{3}{4}\right)^2$$

$$49y^2 - \frac{21}{2}y + \frac{9}{16}$$

6 Difference of two squares

$$(a + b)(a - b) = a^2 - b^2$$

- Reason: $a^2 - ab + ba - b^2$, and the middle terms cancel.
- It turns a product of two numbers equally far from a round number into a subtraction of squares.
- Śrīdharācārya's method for squares** (750 CE): $a^2 = (a + b)(a - b) + b^2$. Choose b so that $a + b$ or $a - b$ is a round number.

Example 6.1 Find 98×102 and 45×55 .

SOLUTION

$$98 \times 102 = (100 - 2)(100 + 2) = 10000 - 4$$

$$45 \times 55 = (50 - 5)(50 + 5) = 2500 - 25$$

9996 and 2475

Example 6.2 Find 197^2 by Śrīdharācārya's method.

SOLUTION

$$197^2 = (197 + 3)(197 - 3) + 3^2 = 200 \times 194 + 9$$

38809

Example 6.3 Write 100 as a difference of two squares.

SOLUTION

$$100 = 50 \times 2 = (26 + 24)(26 - 24)$$

$$26^2 - 24^2$$

7 Patterns explained by identities

- Adding the first two identities: $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$. So twice a sum of two squares is again a sum of two squares, for example $2(5^2 + 3^2) = 8^2 + 2^2$.
- Subtracting them: $(a + b)^2 - (a - b)^2 = 4ab$.
- Three consecutive numbers: $n^2 - (n - 1)(n + 1) = 1$. The square of the middle one is 1 more than the product of the other two.
- $(a - b)(a + b) = a^2 - b^2$, $(a - b)(a^2 + ab + b^2) = a^3 - b^3$, $(a - b)(a^3 + a^2b + ab^2 + b^3) = a^4 - b^4$.

Example 7.1 In a calendar, take any 2×2 block of dates. Show that the two diagonal products always differ by 7.

SOLUTION

Call the dates $a, a + 1, a + 7, a + 8$.

$$(a + 1)(a + 7) - a(a + 8) = a^2 + 8a + 7 - a^2 - 8a$$

7, whatever the value of a

Example 7.2 Show that the square of an odd number is 1 more than a multiple of 8.

SOLUTION

$$(2n + 1)^2 = 4n^2 + 4n + 1 = 4n(n + 1) + 1$$

$n(n + 1)$ is a product of consecutive numbers, so it is even. Hence $4n(n + 1)$ is a multiple of 8.

Proved

8 Many ways to count

- The same pattern can be counted in different ways. Each way gives a different-looking expression, and simplifying shows they are equal.

Example 8.1 A pattern has k rows of $k + 2$ dots at step k . Show that $k(k + 2)$, $(k + 1)^2 - 1$ and $k^2 + 2k$ all describe it, and find the number of dots at step 15.

SOLUTION

$$k(k + 2) = k^2 + 2k \text{ and } (k + 1)^2 - 1 = k^2 + 2k + 1 - 1 = k^2 + 2k$$

All are equal. At step 15: $225 + 30 = 255$ dots.

Example 8.2 Four rectangles, each m by n with $n > m$, are arranged around a square hole to form a large square of side $m + n$. Find the area of the hole in two ways.

SOLUTION

Way 1: large square minus four rectangles, $(m + n)^2 - 4mn$.

Way 2: the hole is a square of side $n - m$, area $(n - m)^2$.

Both equal $m^2 - 2mn + n^2$

9 Common mistakes

Wrong	Right
$(5m + 6n)^2 = 25m^2 + 36n^2$	$25m^2 + 60mn + 36n^2$
$2(x - 1) + 3(x + 4) = 2x - 1 + 3x + 4$	$2x - 2 + 3x + 12 = 5x + 10$
$-3p(-5p + 2q) = -3p + 5p - 2q$	$15p^2 - 6pq$
$(a + 2)(b + 4) = ab + 8$	$ab + 4a + 2b + 8$
$5w^2 + 6w = 11w^2$	Unlike terms; cannot be combined
$3a(2b \times 3c) = 6ab \times 9ac$	$18abc$ (no distribution over multiplication)
$y + 2(y + 2) = (y + 2)^2$	$y + 2y + 4 = 3y + 4$